

All provided solutions should be sent as a publicly accessible repository link.

Problem 1 (mandatory):

The goal is to create a Java train ticketing application. The application should have records of predefined train schedules and train routes consisting of various stations.

The application should be able to support the following functionalities:

- a) ability to book, one or multiple, tickets on trains.
 - it must be taken into account the known bookings for the train so as to prevent overbooking.
 - a confirmation mail should be sent to the customer's given email address after the booking is done.
- b) finding possible departure/arrival times between two stations.
 - if there is no possible link between the stations, an appropriate error message should be shown.
 - the stations could be linked directly or it could be that a changeover between trains would be necessary.
- c) for an administrator, the following additional operations should be possible:
 - ability to add/remove/modify new routes with stations
 - ability to add/remove/modify new trains
 - ability to show what bookings were made for any trains
 - ability to specify which trains have encountered delays. When that happens the trains customers should be notified via email of the delay.

The application should also include a detailed readme file that describes the solution. It should contain examples of the input and output for all supported functionalities.

Problem 2 (optional):

Suggest an interesting solution to a problem you define, and showcase a programatic implementation for that problem.